

CaptionAI: AI-Powered Social Media Caption Generator

Project Description:

CaptionAI harnesses GenerativeAI to provide intelligent social media content generation, offering users a fast and Generative, platform-optimized captions per request, a Hashtag Suggester, which generates ten relevant hashtags blending popular and niche tags, and a Call-to-Action (CTA) Generator, which delivers three punchy, action-oriented phrases tailored to the selected platform and ad type. **Control List Management for ServiceNow**

Utilizing Groq's API with the LLaMA 3.3-70B Versatile model, CaptionAI processes user inputs to deliver fast, high-quality social media content. Built with Flask and deployed publicly via Ngrok, the platform ensures a seamless and user-friendly experience. With secure API key management via environment variables and responsible input handling, CaptionAI empowers marketers, creators, and businesses to craft compelling social media content with confidence.

Scenarios

Date: 29 September 2026

Team members: Devadharshni , vaishnavi, varshini, buvaneshwari, thulasi priya

Scenario 1: A user promoting a B2B SaaS product selects LinkedIn with a Professional tone. CaptionAI generates thought-leadership-style captions with a value-driven narrative, industry-relevant hashtags, and CTAs designed to drive professional engagement such as demo sign-ups or article reads.

Scenario 2: A user running a weekend flash sale selects Instagram with a Promotional tone. The AI generates urgency-driven captions filled with power words, a curated hashtag set targeting deal-seekers, and CTAs like "Shop before it's gone!" to maximize conversion.

Technical Architecture

Project Workflow:

Milestone 1: Model Selection and Architecture

- **Activity 1.1:** Generate a Groq API key and configure it securely in the project environment.
- **Activity 1.2:** Research and select the appropriate generative AI model from Groq for social media content generation.
- **Activity 1.3:** Define the architecture of the application, detailing interactions between the frontend, backend, and AI integration.
- **Activity 1.4:** Set up the development environment, installing necessary libraries and dependencies for Flask and the Groq API.

Pre-requisites

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop the core functionalities: Caption Generator, Hashtag Suggester, and Call-to-Action Generator.
- **Activity 2.2:** Implement the Flask backend to manage routing and user input processing, ensuring smooth API interactions.

Milestone 3: App.py Development

- **Activity 3.1:** Write the main application logic in app.py, establishing routes for each feature and integrating AI responses.

Milestone 4: Frontend Development

- **Activity 4.1:** Design and develop the user interface using HTML, CSS, and JavaScript, ensuring a responsive and intuitive layout.
- **Activity 4.2:** Create dynamic templates with Flask's render_template to display results based on user interactions.

Milestone 5:

- **Activity 5.1:** Prepare the application for local deployment by configuring the server environment and ensuring all dependencies are correctly installed.
- **Activity 5.2:** Deploy the application publicly using Ngrok to expose the local Flask server over a secure public URL.

Key Accomplishments:

- Reduced ACL configuration time by over 70% through automated script execution.
- Eliminated manual misconfigurations with real-time pre-execution validation.
- Provided 100% audit compliance with automated logging and one-click rollback.

Team Contributions & Sign-off

submission date: 02 october 2026

Team id: 6ab63861dcdae33acb394aaf

PROJECT TITLE: Script-Controlled ACL – Restrict Record Access Based on Field Value

Team Member Responsibilities:

1. Devadharshni M(Team Leader):

- Architecture design, core ACL script engine, team coordination, and project reporting.

2 .Varshini A

- Systemarchitecture modeling,RESTAPlintegration,andperformancebenchmarking.

3. Bhuvaneshwari K

- Frontend Service Portal development, script editor UI, and syntax validation.

4. Vaishnavi V:

- Bulk operations module, rollback snapshot mechanism, and automated test cases.

5. Thulasi priya P

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